

DS-3001

Data Analysis and Visualization

Data Preparation & EDA

Data Cleaning vs Preprocessing

- Data Cleaning
 - Modifying raw data to be unambiguous, well-formatted, and readable by computers without altering original files.
 - Goal: Correcting errors, standardizing column names, fixing variable types, and eliminating ambiguities.

Data Cleaning vs Preprocessing

- Data Cleaning
 - There are many different possible clean versions of every dataset
 - Document judgment calls you made
 - Investigate whether your downstream results are stable.

Data Cleaning vs Preprocessing

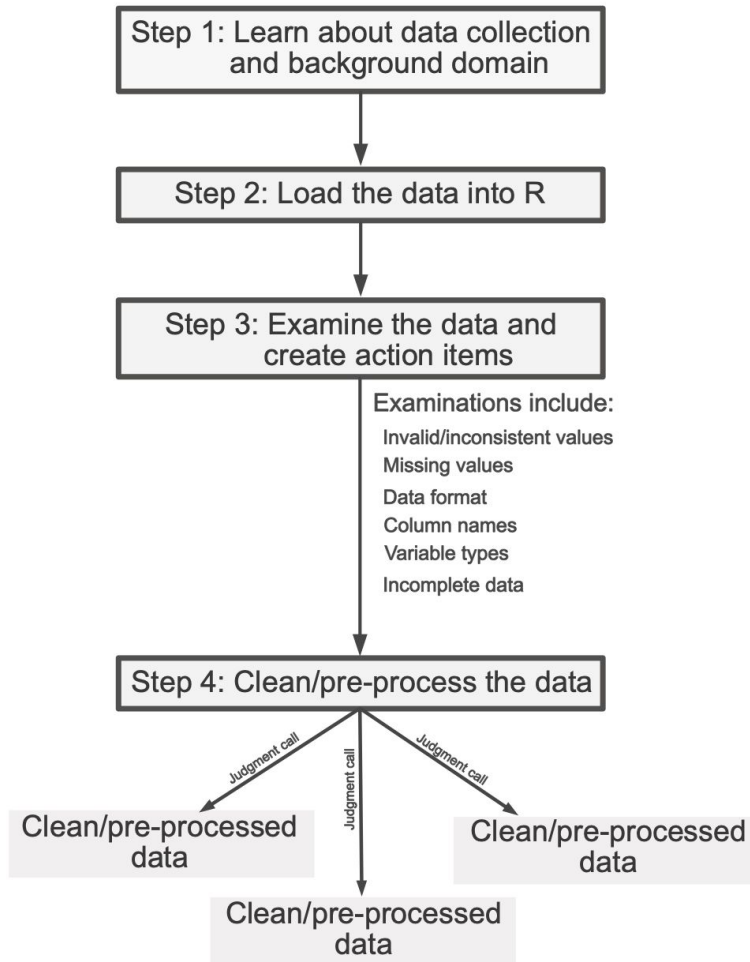
- Data Preprocessing
 - Modifying clean data to meet specific algorithmic requirement (e.g., scale normalization, numeric encodings, featurization).
 - Goal: Preparing data objects specifically tailored for model execution and statistical analysis.

Data Cleaning vs Preprocessing

- Data Preprocessing
 - There are many different possible preprocessing outputs
 - Document judgment calls you made
 - Investigate whether your downstream results are stable.

Why Data Cleaning Matters

- Time-Intensive Bedrock
 - Data preparation consumes significant slice of the total time spent on any data science project lifecycle.
- Requires Human Judgment
 - Cannot be fully automated. Requires human critical thinking, domain expertise, and contextual decision-making.
- Deep Domain Learning
 - Hands-on cleaning builds intimate knowledge of data limits, quirks, and collection flaws essential for valid downstream results.



The 4-Step Procedure

- Domain Research
 - Learn collection details, survey parameters, and domain constraints.
- Load Data
 - Import safely without modifying the raw source files directly.
- Examine & Plan
 - Identify messy traits and formulate precise action items.
- Execute Clean
 - Apply changes in encapsulated, reusable functions.

Reviewing Background and Domain Information

- What does each variable measure?
- How was the data collected?
- What are the observational units?
- Is the data relevant to my project?
- What questions do I have, and what assumptions am I making?

Loading Your Data

- How your data is being stored?
- If your data contains multiple tables, is there a key variable that connects them?
 - Are there multiple entities in your data e.g. in a deeply nested json?
- What parts of the data are relevant to the question being asked?

Examine the Data and Create Action Items

- Invalid or inconsistent values.
- Improperly formatted missing values.
- Nonstandard data format. (“tidy” vs “long” format)
- Messy column names.
- Improper variable types.
- Incomplete data.
- Addressing Any Remaining Data Specific Questions and Assumptions

Additional Common Preprocessing Steps

- **Standardizing or normalizing variables** that are on different scales so they are comparable (see sec-comparability in sec-eda)
- **Transforming variables** so they follow a more Gaussian (“normal” bell curve) distribution (see sec-pca-transformation in sec-dimension-reduction)
- **Splitting categorical variables** into binary variables, called “one-hot encoding” in ML
- **Featurization** (creating new variables from the existing variables)

Best Practices

- **Never overwrite raw data:** Always write code to transform memory-loaded objects, keeping original source files untouched.
- **Encapsulate in functions:** Wrap cleaning steps in modular functions with explicit parameters for key judgment calls.
- **Document alternative choices:** Record all subjective decisions in DSLC documentation to enable stability analysis.
- **Investigate stability (PCS):** Test whether downstream conclusions hold across different valid cleaning options.

References

1. Veridical Data Science by Bin Yu and Rebecca L. Barter - Chapter 4
 - <https://vdsbook.com/>